

Sciencetech 9055 / 9055F, 9057 / 9057F

USB Controlled

Direct Drive Monochromator

User Manual



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1 Introduction

The Sciencetech 9055 / 9057 is a modified Czerny-Turner monochromator, with a USB stepper motor driving the grating turret directly through a worm-gear drive. The gratings in the 9055 / 9057 can be changed automatically with a triple grating turret (double through grating turret for F version) the same drive that does the scanning during operation. The design of this instrument allows a relatively high resolution for an instrument of this size throughout the spectral range.

The Sciencetech 9055 / 9057 can be controlled from any computer with a USB 2.0 port. The automatic scanning is done with a USB stepping motor controlled by an integrated internal microprocessor and the appropriate software such as SciSpec. Communication of the microprocessor with the host computer is through a USB 2.0 interface.

The Sciencetech 9055 / 9057 can operate in the following modes:

1. Scanning monochromator driven by USB stepping motor system and controlled from a host compatible computer
2. Flat-field spectrograph for use with linear diode array detector systems

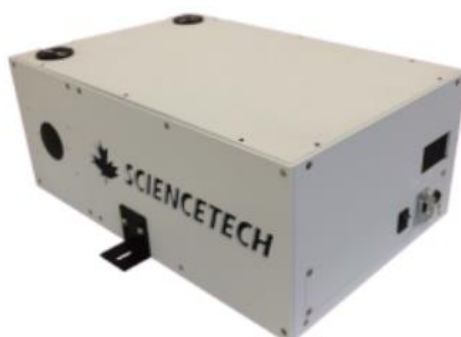


Figure 1. Sciencetech Model 9055

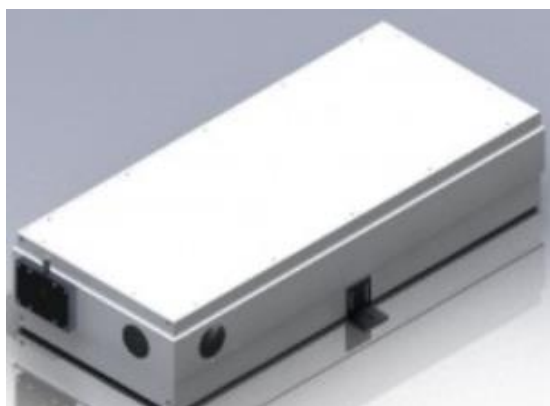


Figure 2. Sciencetech Model 9057

2 Safety

2.1 Warning – Crushing Hands



If instructed to open the monochromator, make sure to turn off and power down the monochromator. The moving turret can pose danger of crushing of hands.

2.2 Do not stare AT light source



If you intend to integrate or use the monochromator with a light source, do not stare or directly look at its output slit. Bright light can exit the output slit.

2.3 Eye Protection



It is recommended when operating the monochromator with a light source to use the appropriate goggles with respect to the light source that is being used.

3 Unpacking, Inspection and Installation

3.1 Unpacking and Inspection

The instrument is packaged in such a way as to minimize damage during transport. If the package is damaged or if after unpacking any signs of damage become apparent, a claim should be filed with the carrier immediately.

If the instrument must be returned, contact Sciencetech Inc. for approval prior to shipping. A full description of the reason for return should be included.

First, inspect the exterior of the monochromator for any noticeable defects. If any are present, contact Sciencetech Inc. immediately.

ONLY if damage is suspect may it be necessary to inspect the inside of the monochromator to make sure the mirrors and other components are in their proper configurations. The level of stresses the shipping carton(s) experience during transit is out of Sciencetech's control. There are times when a particularly harsh impact event can unseat one or both mirrors on their holders which will render the instrument completely out of alignment. This is easily remedied, however by following steps now outlined.

- (i) In a clean, dust-free environment, carefully remove the 10 x M3 cap head screws that attach the lid to the monochromator body. Note their positions for when the lid is to be re-attached. Do NOT overtighten the screws or the holes will strip.
- (ii) If there is a flipper mirror(s), loosen the set screw that holds the turning knob onto the shaft and remove it.
- (iii) Carefully remove the lid. It is advisable to wear a surgical mask so as to not breath on the delicate grating(s) and mirrors.
- (iv) Inspect the mirrors. Each one should be oriented on their holders as shown in the figure below. Note that the three set screw tips on the mirror holder should be seated in the corresponding three holes in the mirror backing plate.

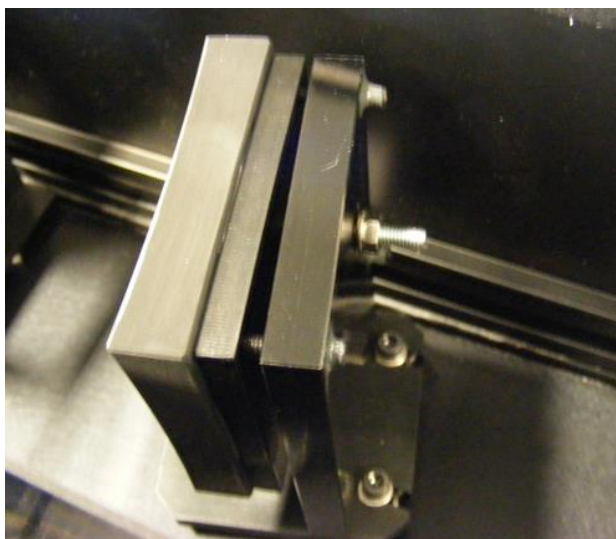


Figure 3. Mirror holder with set screws

If one or both mirrors appear as shown in the figure below, their backing plate must be re-seated on the three set screw tips on the mirror holder. Note that there is only one position in which this can occur.

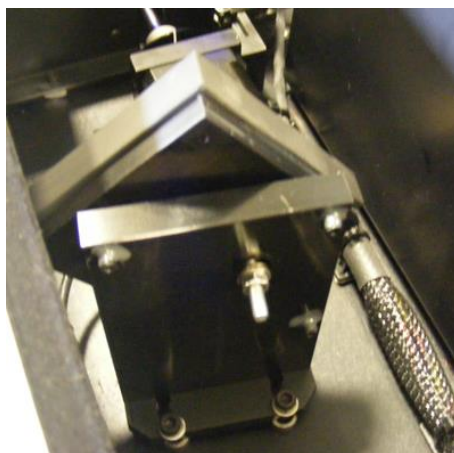


Figure 4. Mirror needing re-seating.

Wearing gloves, simply push in on the PEM bolt with the spring-loaded nut and simultaneously pull the mirror away from the holder and rotate it so that it seats on the three set screw tips.

While the lid is off the monochromator, it is a good idea to check the wiring at the rear inside panel. Make sure the USB connector is firmly in place and the connectors on the small PCB with the relay are not loose. See the figure at left.

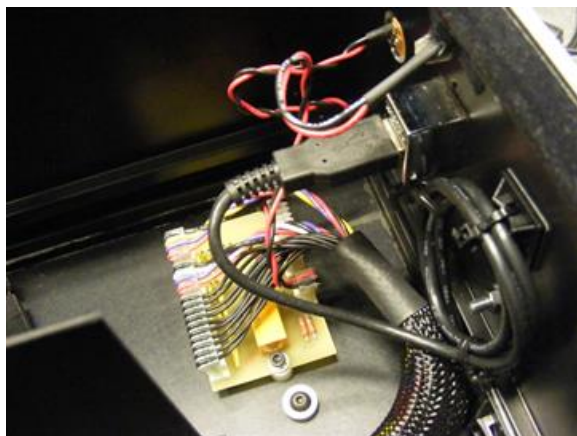


Figure 5. Wiring of rear inside panel of monochromator.

3.2 Included Components

The packing box should contain the following components:

- Sciencetech 9055 / 9057 monochromator
- 24VDC output power supply, 100-240 VAC input
- Mounted entrance slit, and any optional slits
- Brackets for attachment to an optical table
- USB 12- or 16-bit computer board for data acquisition (optional)
- Type B to Type A USB Cable
- SciSpec (optional) or SciMono (included) software disk

3.3 Description of connections/controls

There are three main connection/button on the rear panel of 90XX series monochromators (Figure 6).

1. Rocker switch that allows user to power up the motorized turret.
2. USB B-A panel adaptor, which allows user to establish signal communication between SciMono/SciSpec/SciPV software on PC with the monochromator through USB B-A cable.
3. 24 VDC power jack that provide power to the monochromator from the 24 VDC power brick supplied with the system.

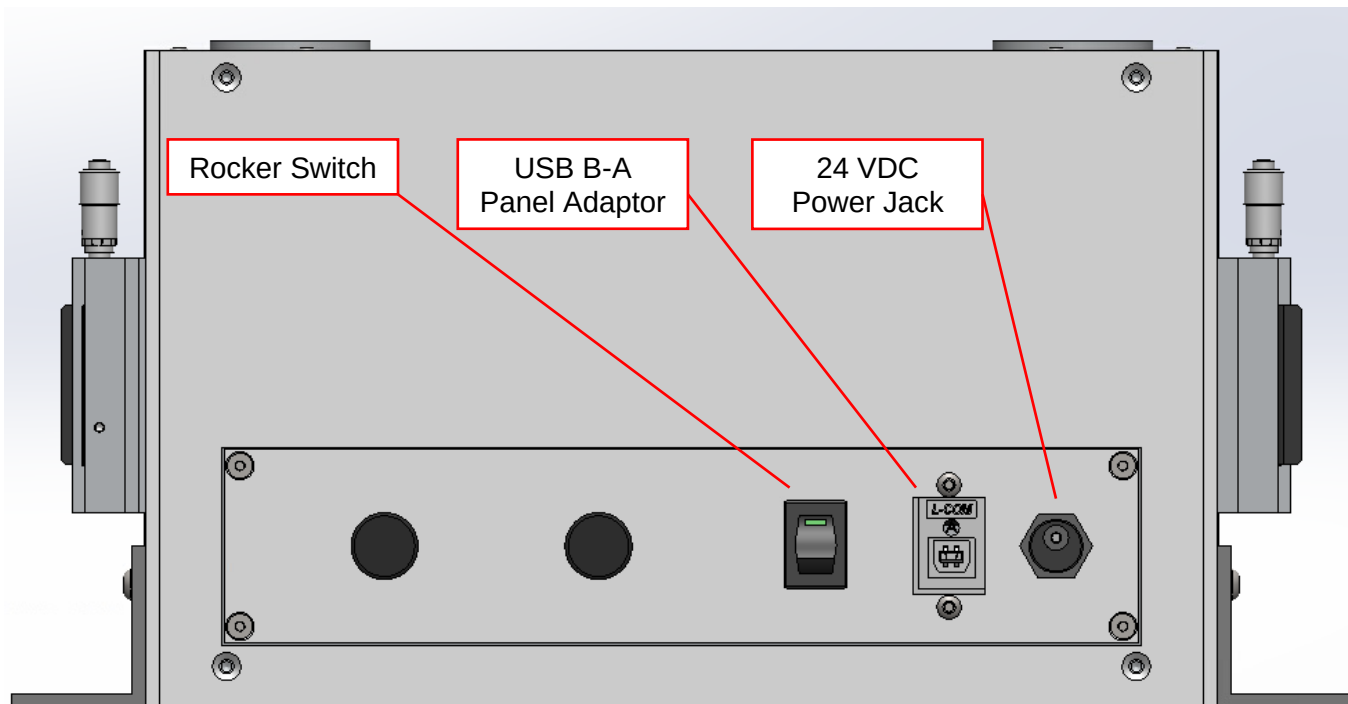


Figure 6. Connections and control on the rear panel of 90XX series monochromators.

3.4 Installation Procedure

NOTE: The safety of any system incorporating this equipment is the responsibility of the assembler of the system.

1. Place the monochromator in a desirable position. The monochromator should be installed on an optical table. The monochromator has a specially designed base that allows it to be used on any surface. However, the monochromator is a precision optical instrument and it must be used in an environment that minimizes as much as possible any vibration.
2. After Ensuring that the power switch is off, connect the 24 VDC power supply male connector to the female connector on the back panel of the monochromator. Then plug in the power supply to a 100 – 240 VAC outlet. NOTE: The monochromator should be powered using a power supply rated for not more than 65W. The power supply provides the overcurrent protection for the instrument.
3. Once mounted, copy the SciSpec or SciUSBMono software from the USB key to your desktop. Install the software according to their instructions; refer to the SciSpec User Manual for the optional SciSpec software. Connect the monochromator to the control computer and switch it on.

4 Monochromator Features

4.1 General Features

The basic design is a modified Czerny-Turner monochromator.

The 9055 / 9057 has no sine arm and can only be operated in computer controlled automatic scanning mode using the proper drive software. They also have neither manual control nor mechanical wavelength display.

The entrance and exit slits can be mounted on the front or sides of the monochromator (see diagram below). They are mounted on the front ports, unless the customer has requested otherwise. As an option, 45° flipping mirrors can be installed in either or both of the entrance and exit ports in order to access both the front and side slits. The manual-switching knobs for the 45° flipping mirrors are located on the top of the monochromator (see diagram below).

A detector can be mounted on either or both exit ports. If one detector is mounted on each exit port the 45° flipping mirror can be used to manually select between the two detectors. Automatic computerized detector switching can be performed with Sciencetech's dual detector box accessory (see section 8: Options and Accessories).

The 9055 / 9057 can be shipped with up to 3 gratings (2 gratings for the F version) mounted according to the requested specifications.

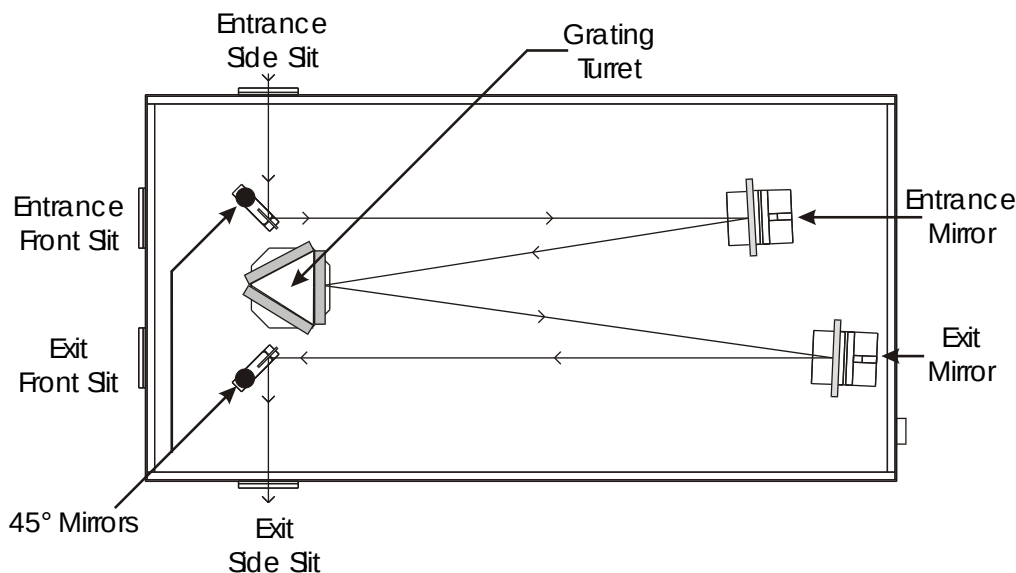


Figure 7. Model 9055 with entrance and exit mirror labeled.

Note: for the 9057 and 9057F, the Entrance and Exit Mirror distances are symmetric.

Flat field tilt for diode arrays: -1.69°

Model 9057: All distances are referred to entrance slit; see Figure 8

Entrance mirror (mirror 'A')

Mirror size = 76.2mm diameter
(9057F/ 9057F)
 Z_A = 457.2 mm (focal length)
 X_A = 0 mm
Tilt angle A = **5.47°**

Exit mirror (mirror 'B')

Mirror size = 76.2mm diameter
(9057 / 9057F)
 Z_B = 457.2 mm (focal length)
 X_B = 140 mm
Tilt angle B = **4.73°**

Grating

Grating size = 50 x 50 mm² (9057)
= 64 x 64 mm² (9057F)
 Z_G = 67.5 mm
 X_G = 75 mm
Grating angle = 0° (at 0 nm)

Exit slit

Z_E = 0 mm
 X_E = 140 mm

Flat field tilt for diode arrays: – 1.40°

4.3 Special layout

Most competitor's instruments that specify resolutions of the order or better than 0.2 nm (spherical aberration limit) show significant stray light. This light is produced by multiple reflections in the grating (a result of their very small off-axis angles) and by insufficient internal baffling (for the same reason that imposes inner space constraints). Some competitor's instruments also have vignetting for the same reason. The two different focal lengths in Sciencetech's design permit high throughput and resolution while maintaining the off-axis angles large enough to avoid vignetting and to minimize stray light.

4.4 Optics

Light focused at the entrance slit is collimated by the entrance spherical mirror and reflected towards the grating. At the grating, the beam is diffracted and depending upon the grating angle, collimated beams of the required wavelength are sent to the exit mirror. The exit mirror focuses the light in the exit slit. Only wavelengths within a given bandwidth can exit. The limit of the optical resolution and the width of the slits determine the bandwidth.

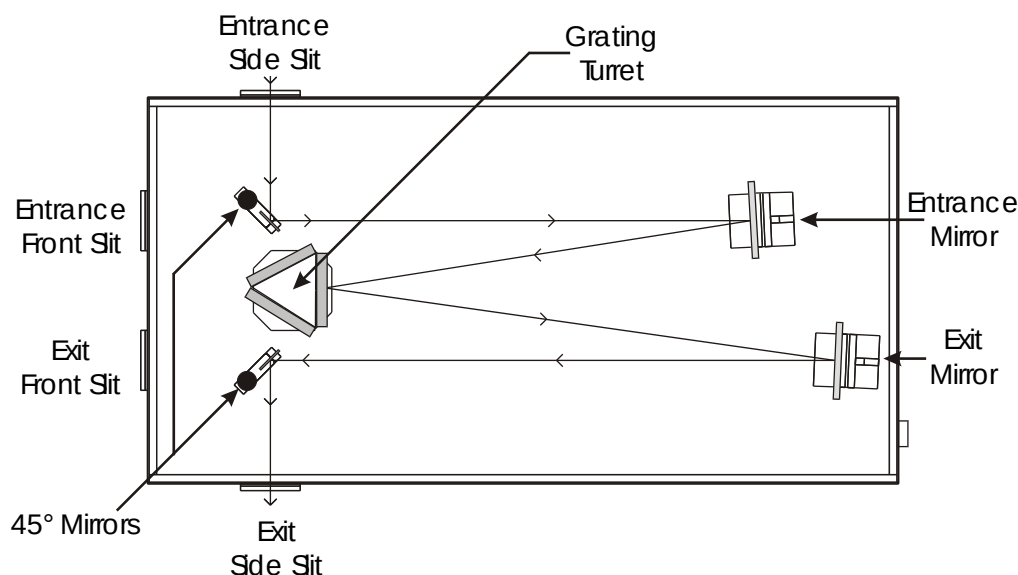


Figure 9. Diagram 9055 labeled with entrance, exit, and mirrors at 45.

Front or side entrance and exit slits are available. The side slits are accessed through 45° plane mirrors. These plane mirrors do not degrade the image quality when they reflect the beam. Consequently, the resolution and calibration are equally good regardless of which optical path is chosen. Versions with two entrance or exit slits use flipping mirrors. Diode arrays of up to 25.4 mm wide can be mounted on the front exit port.

4.5 Input and Output Ports

Two identical flanges on the input ports and two on the output ports of the instrument allow side or front slits to be mounted at the input or output of the monochromator. The side input flange (at 90°) is typically covered; the front input flange is the operational one and is fitted with a slit assembly. If the side input port option is ordered, a fixed 45° mirror is installed in such a way that the beam enters and exits through the side ports keeping the entrance slit in line with the exit slit and the detector. For either port (front or side) the central ray is perpendicular to the face of the instrument.

Either port can be used as the input or output port except when detector arrays are used. The port used must be based on the numerical aperture of the source or detector. The element with the higher numerical aperture should be located in the left port (labeled input). Bolt patterns of the port specified in the slit section.

Array detectors must be connected to the front or side output port with an adapter with the appropriate tilt to match the flat field. Up to 1 inch diode arrays (1024 elements, 25 µm pitch) can be used.

4.6 Selectable Gratings

As a general rule (except with a diode array detector), if high resolution is desired, use the grating with the maximum possible number of lines that can read the desired wavelength.

Table 1. Wavelengths for gratings of high resolution.

Wavelength	Recommended Grating for High Resolution [l/mm]
< 400 nm	3600
< 600 nm	2400
< 1.2 µm	1200
< 2.4 µm	600
< 10 µm	150
< 30 µm	50

Up to 3 gratings (2 for the 9055F) can be installed in the 9055 monochromators. If the spectrometer has been purchased with more than one grating the computer can select the grating. When only one grating is purchased it is mounted in position 1.

4.7 Filter Wheel (optional)

Sciencetech filter wheels can hold six, eight or twelve, 1-inch round or square filters up to 5 mm thick. One position is usually supplied with a dark cover for zero referencing. Second, third or higher order wavelengths of light diffracted from the grating can be eliminated by the use of a long pass filter.

Filter wheels can be manual or automatic. Automatic filter wheels are computer controlled, allowing the computer to switch filters during a wavelength scan to better isolate the desired wavelength. If you have purchased a filter wheel, see the filter wheel user manual for more information. If you have purchased Sciencetech software for controlling the filter, consult the manual for more information.

5 Principles of Operation

5.1 Dispersion

For the 9055 the dispersion vs. line density on the grating can be approximated by:

$$D_{9055} = 5000 / (\text{grating lines per mm})$$

$$D_{9055F} = 3960 / (\text{grating lines per mm})$$

giving: $D_{9055} = 8 \text{ nm/mm}$ and $D_{9055F} = 6.6$ with a 600 l/mm grating
 $D_{9055} = 4 \text{ nm/mm}$ and $D_{9055F} = 3.3$ with a 1200 l/mm grating

The dispersion is approximately constant over the entire wavelength range.

5.2 Resolution

Resolution is the most important characteristic of a spectrograph and is used as a measure of an instrument's ability to separate adjacent spectral lines. Two peaks are considered resolved if the distance between them is such that the maximum of one falls on the first minimum of the other. This is called the Rayleigh criterion. This depends on many factors such as the width of the gratings, focal length of the instrument, system aberration, entrance and exit slit widths.

The theoretical resolution depends on the width of the entrance and exit slits. It is the width of the narrowest bandpass δ_λ that can be resolved at wavelength λ , which is given by:

$$\delta_\lambda = (W)(D)$$

where:

W	=	width of the slits (see below)
D	=	reciprocal linear dispersion.

There is a minimum value δ_λ even when the slit is almost closed. This value is determined by the optical characteristics of the system and is limited by optical aberrations. This value is generally given as the theoretical resolution of the instrument. For the 9055 with a 1200 lines/mm grating, $\delta_\lambda = 0.2$.

5.3 Resolving Power

Resolving power depends on the width of the gratings and can be estimated by this equation:

$$R = \lambda / \delta_\lambda$$

For a diffraction grating, the theoretical value of the resolving power is

$$R = (m)(N)$$

where:

R	=	resolving power
λ	=	the central wavelength of the spectral line to be resolved
δ_λ	=	resolution
m	=	diffraction order
N	=	total number of lines engraved on the total area of the grating

5.4 Optic Equations for Czerny-Turner Monochromators

For a Czerny-Turner monochromator with entrance angle = $2A$ and exit angle $2B$, the angle between the incident and diffracted beam in the grating is $2A + 2B = 2\Phi$.

The grating equation is:

$$m \lambda = d [\sin(2A + \Theta) - \sin(2B - \Theta)]$$

where:

m	=	diffraction order
λ	=	wavelength
d	=	separation between grooves in the grating
Θ	=	grating angle with respect to central monochromator axis
$(2A + \Theta)$	=	incident angle with respect to the grating normal
$(2B - \Theta)$	=	diffracted angle with respect to the grating normal

The grating equation can be rewritten as:

$$m \lambda = 2 d \cos(\Phi) \sin(\Theta + \Theta_0)$$

where:

Φ	=	$A + B$
Θ_0	=	$A - B$

For the specular reflection (zero order diffraction) to exit at the required position, the grating has to be at an angle $-\Theta_0$. In a symmetric monochromator $\Phi = 2A$ and $\Theta_0 = 0$. The 9055 is an asymmetric monochromator ($\Theta \neq 0$).

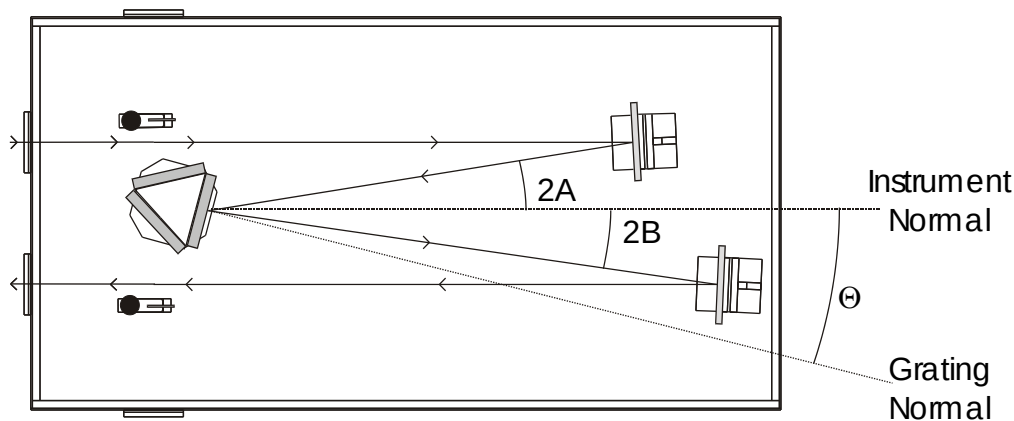


Figure 10. Zero order diffraction and positions.

Values for the 905X monochromators:

$A = 15.40^\circ$ (9055, 9055F); $A = 5.47^\circ$ (9057, 9057F)
 $B = 10.01^\circ$ (9055, 9055F); $B = 4.73^\circ$ (9057, 9057F)

5.5 Aberration Correction

The Sciencetech 9055 monochromator uses input and output spherical mirrors of different focal lengths and different spherical mirror incidence angles to eliminate coma at 600 nm. The resolution is limited by the spherical aberration and, for tall slit openings, by the slit curvature.

5.6 Blazed Grating Efficiency

Blazed diffraction gratings have grooves cut or "blazed" at a particular angle so that most of the light is diffracted at a particular wavelength for a given diffraction order. Typically, the grooves are flat but form an angle with the grating surface. The peak efficiency of a blazed grating corresponds to diffraction angles close to the angles for specular reflection on the groove surface. The blaze wavelength is the wavelength at which the grating has maximum efficiency for first order diffraction. When operating in a higher order m , the peak efficiency will be at the blaze wavelength divided by m .

To estimate the useful range of a grating, the "2/3–3/2" rule can be used. The efficiency will drop to half of its peak value at about 2/3 of the blaze wavelength and again at 3/2 of the blaze wavelength. The drop is very sharp for wavelengths $< 2/3$ of the blaze, so it is not recommended its use below this value. The drop at long wavelengths is slow, so generally the grating is useful above 3/2 of the blaze limit. But because the s and p components have different cutoff wavelengths, the exit radiation will be strongly polarized.

The discussion above is valid for ruled or holographic gratings that are "blazed". Standard holographic gratings have grooves with a sinusoidal profile. For these gratings the efficiency is more or less constant over the nominal working range. Standard efficiency for these gratings is approximately 30%, except for the "high modulation" gratings where the efficiency for large incidence angles can be as high as 60%.

As a general rule (except with a diode array detector), if high resolution is desired, use the grating with the maximum possible number of lines that can read the desired wavelength.

Table 2. Grating with high resolution based on wavelength.

Wavelength	Recommended Grating for High Resolution [l/mm]
$< 400 \text{ nm}$	3600
$< 600 \text{ nm}$	2400
$< 1.2 \text{ }\mu\text{m}$	1200
$< 2.4 \text{ }\mu\text{m}$	600
$< 10 \text{ }\mu\text{m}$	150
$< 30 \text{ }\mu\text{m}$	50

6 Operating instructions

NOTE: if this equipment is used in a manner not specified by Sciencetech, the protection provided by the equipment may be impaired.

6.1 Introduction

The Sciencetech 9055 / 9057 can be controlled from any computer with a USB 2.0 port. The automatic scanning is done with a USB stepping motor controlled by an integrated microprocessor. Communication of the microprocessor with the host computer is through a USB 2.0 interface. Data acquisition is performed using an external USB computer interface board (optional). This allows software like SciSpec to coordinate scanning and data acquisition to acquire spectral data and plot it.

6.2 Slits

The slit width can be varied from 10 μm to 5 mm and is controlled by the micrometer screw on top of the slit. The micrometer divisions are in 10 μm and the slit width corresponds to the reading.

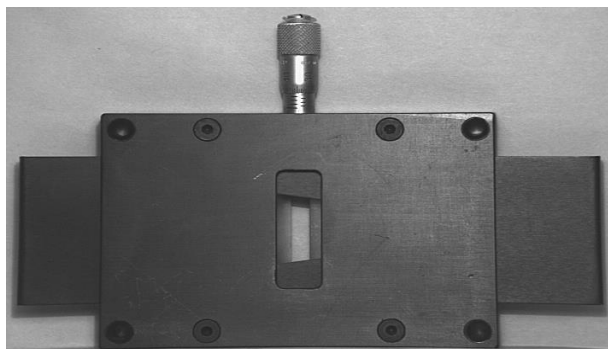


Figure 11. Slit located in center of image.

The slit height is varied manually with a slide. Available slides vary from 0 to 30 mm in height. There are 4 heights of vertical curtains available: 0 to 5 mm, 0 to 10 mm; 10 to 20 mm and 20 to 30 mm. Each slit comes with a set of 4 vertical curtain sliders. To change a slider, simply grab it by the end with a small bend upwards in it and pull it out. Then insert the new slider in the same manner. Note that the slit should be closed when this operation is performed. See Figure 14 below.

The slit is fastened to the input/output port by the four outside screws (see Figure 15). The four inner screws are used to mount auxiliary equipment (i.e. photomultiplier housing) to the slit.



Figure 12. Exchanging a vertical curtain slider; 20 mm to 30 mm shown.

The slit bolt pattern is depicted below. There are 8 threaded M3 holes on the slit. The “outside” 4 holes are for mounting the slits to the monochromator. Also, the same holes can be used for direct mounting a linear diode array. The “inside” 4 holes have been made on a slit for mounting input/output devices such as a detector.

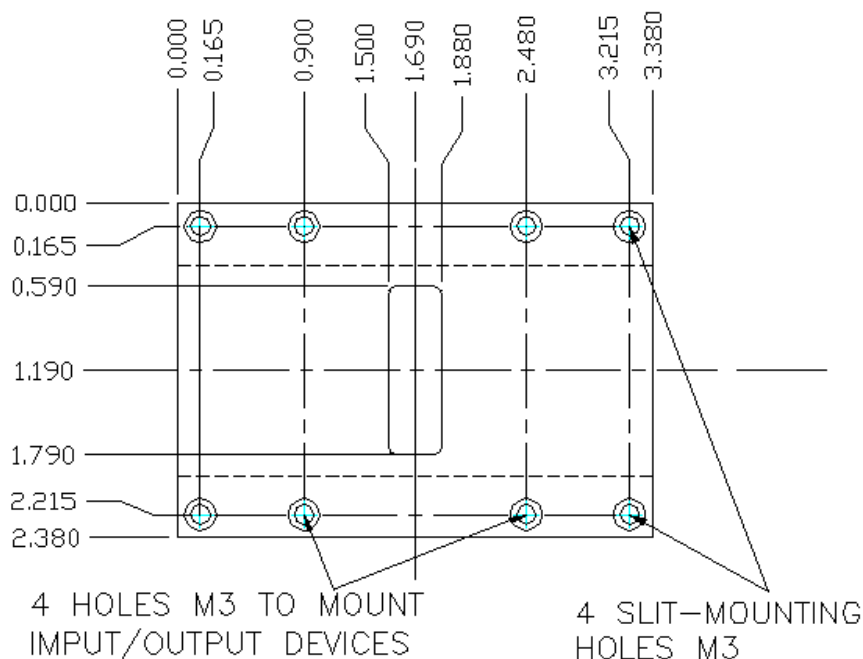


Figure 13. Diagram of slit bolt pattern.

6.3 Grating Selection

If the spectrometer has been purchased with more than one grating the control software can select the grating that is to be used. The standard grating order is in descending pitch (ascending wavelength ranges). When only one grating is purchased, it is mounted in position 1. See section 11 for a list of available gratings.

6.4 Installing and aligning a grating

If you have purchased a Sciencetech Monochromator, its grating(s) will have come pre-installed and pre-aligned with your system. Should you wish to change or upgrade your desired spectrum and have purchased extra gratings, you may need to install them in your monochromator. Follow the instructions below to align and calibrate your grating.

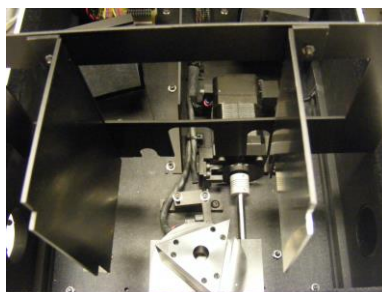


Figure 14. Top view of monochromator.

1. Remove the baffle surrounding the grating by removing the two screws holding it on by the base with a 2.5mm hex key.
2. Remove one of the side hole plugs by hand so that a laser beam can be sent into the monochromator from the side.

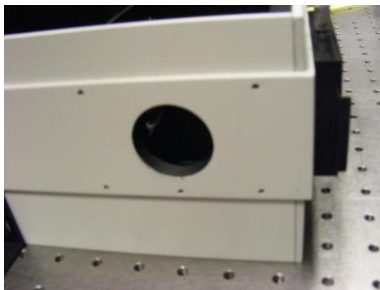


Figure 15. Hole to enter laser beam

3. Your grating will come with a protective plastic cover. Wearing powder-free latex gloves, carefully remove the cover of the grating. **WARNING** Do not touch the surface of the grating with anything.

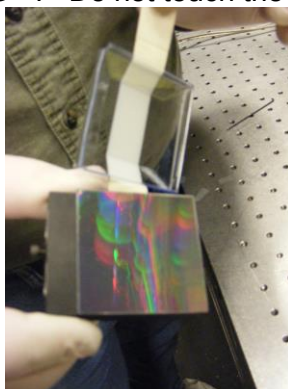


Figure 16. Grating with protective cover



Figure 17. Collimated laser.

4. Set up a collimated laser at a height of 3" and ensure that it is level by using two irises set to the same height, spaced across a table.

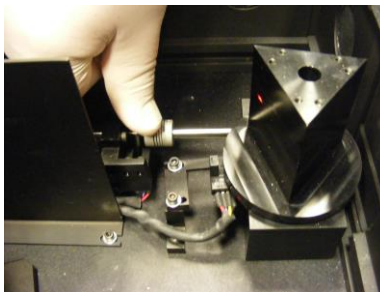


Figure 18. Manually rotating grating drive.

5. Ensuring all electrical connections are unplugged from the monochromator, manually rotate the grating drive to bring the desired grating slot into a position parallel to the input beam of the laser.

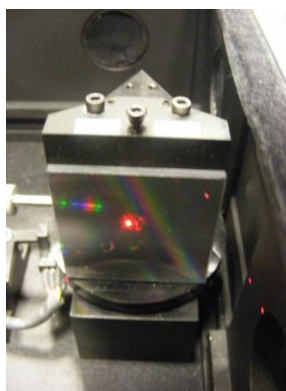


Figure 19. Installed grating with screws and laser beam centered.

6. Install the grating into the vacant slot by gently holding it up against the holder. With the two provided screws, secure the grating into position by tightening into place with a 2.5mm hex key. Ensure that the provided springs are in place. Move the monochromator into place such that the laser beam hits roughly the center of the grating.

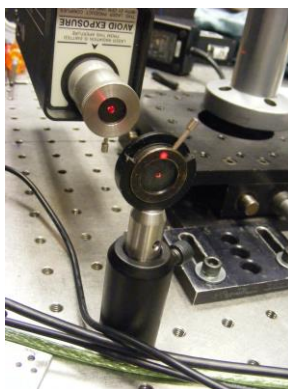


Figure 20. Locating reflections of laser line on iris surface.

7. By manually rotating the grating drive from step 5, locate the reflection of the zero and higher orders of the laser line on the surface of the iris. If the grating is not level, the order lines will appear as though they are higher or lower than the iris opening.

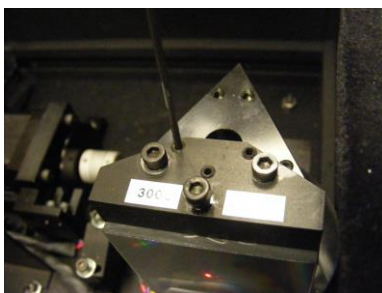


Figure 21. Adjusting three set screws to level grating.

8. To level the grating, adjust the three set screws on the grating holder to adjust the grating back and forth until the reflected laser beam passes through the iris. When level, the laser line will pass through the iris both on the way to and from the monochromator, (see picture from step 4).

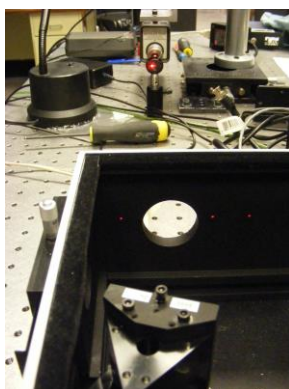


Figure 22. Complete set up with adjusting y-position.

Repeat steps 7 and 8 for higher orders of the laser. By continually rotating the grating drive, you will notice the orders passing by the iris at different times. Remember, only the y-position (height) of the beam is necessary to align with the iris opening. It may be necessary to return to previous orders to ensure they are still leveled.

9. When the grating is aligned, re-install the side hole plug and the grating baffle. Refer to the SciSpec USB Manual for instructions for calibrating the grating. If you have any questions, please contact Sciencetech customer support.

6.5 Flipping Mirrors

This is an upgrade to many of Sciencetech's "Standard" line of monochromators to allow the use of a second input (if DPIN-ST-MAN is purchased) port or a second output port (if DPOUT-ST-MAN was purchased).

The heart of this functionality is a manual flipping mirror that can be installed at the input, output or both ports of the monochromator. This provides the possibility to select between different input and output ports as shown in Figure 23 below:

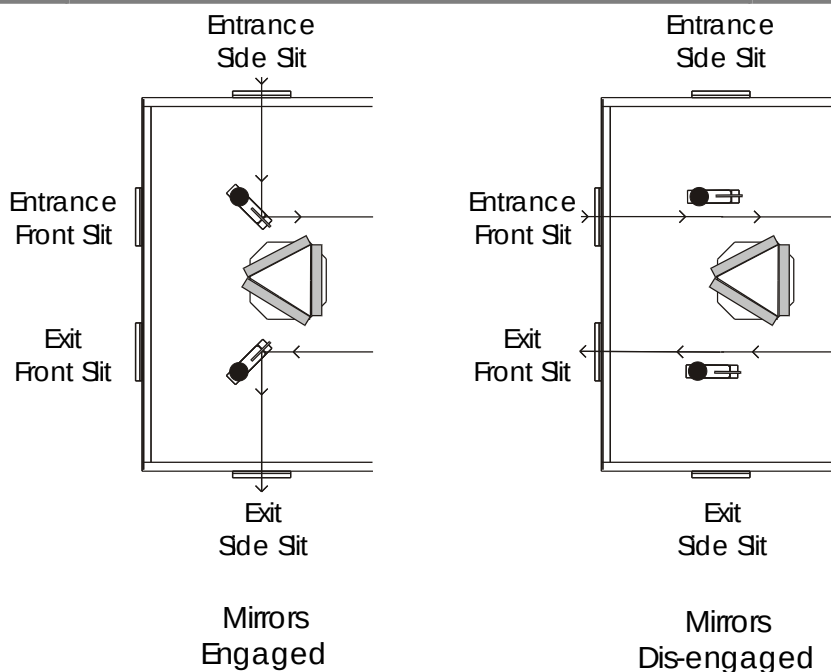


Figure 23. Diagram with different mirror orientations.

To turn the mirror from one position to the other (e.g. from side input (SIDE) to front input (FRONT) in Figure 24 below) follow these steps:



Figure 24: Grasp the input flipper knob and push it all the way down.



Figure 25. Turn the knob until the arrow on the knob align with the FRONT arrow on the label.



Figure 26. Keep a firm grip on the knob and slowly bring it up to the lock position. **DON'T** let it snap back!

6.6 Mirror Care

All mirrors used in Sciencetech monochromators are first surface, aluminized and overcoated for protection and improved reflectance.

< ! > WARNING < ! >

Do not touch the mirror surfaces. Remove dust with a gaseous lens cleaner (not wiping or rubbing).

If the mirrors become dirty (grease, fingerprints, water spots), the cleaning procedure depends on the type of mirror coating:

Visible enhanced mirrors (typically SiO₂ coating) – clean with lens cleaning liquid and tissue. **Do not rub**: this can produce micro-scratches that will degrade their performance.

UV enhanced mirrors – clean chemically, any mechanical cleaning will degrade their performance. Use clean room detergent followed by deionized (DI) water rinse and blow dry with clean air. Alternatively: acetone, alcohol, DI, blow dry.

If the monochromator is modified to use the side ports, two flat mirrors are used for path selection. The removal and later replacement of these 45° flat mirrors may produce changes in the accuracy of the instrument. Repositioning of these mirrors is difficult and should not be attempted by an inexperienced user since re-alignment of the system may be necessary. If the user wants both ports readily available swing-away mirrors are an option.

The input spherical mirror is used to collimate the radiation coming from the entrance slit. The output spherical mirror focuses the radiation in the exit slit. These mirrors are factory adjusted for maximum resolution. Any misalignment would degrade the performance of the system.

< ! > WARNING < ! >

Do not turn the alignment screws in the back of either spherical mirror. This will degrade alignment and calibration.

6.7 Spectrograph Mode

The 9055 / 9057 can be converted into a spectrograph for diode arrays of up to 1 inch in width. In this case an optional diode array mount, tilted at an angle that optimizes the focusing for the different wavelengths, replaces the front exit slit. Sciencetech offers a linear diode array detector system with optical analysis software.

7 Electronics

The heart of the system is the Stepper Motor with integrated driver and controller. The stepper motor controller is a very compact unit that mounts to the back of the motor. This motor communicates via a USB 2.0 interface. Opto sensors are used to indicate to the motor controller when an end switch condition has been reached; for example, when homing the grating turret drive. The motor controller also allows a relay to turn the opto sensors off when not homing the grating so that light emitted by the sensors does not interfere with monochromator measurements.

8 Options and Accessories

8.1 Side ports

Flat mirrors at a 45° angle deflect the input and/or output beams 90°. In this way it is possible to have in-line input and output.

8.2 Flipping mirrors

Swing-away mirrors can replace one or both of the standard fixed flat mirrors, which select the 90° ports. These mirrors provide a mechanism for easily switching between front and side entrance ports without risking misalignment of the system. The swing-away mirrors are controllable from the outside of the monochromator suitable for use with a dual detector or dual light source setups. See Section 6.5. Motorized control is available.

8.3 Motorized Slits

Sciencetech motorized slits are high precision straight bilaterally adjustable slits. Width adjustment is driven by a stepper motor and can operate in manual or automatic mode.

8.4 Filter wheel

A filter wheel can be added to resolve the sorting order of the grating. Sciencetech filter wheels can hold 6, 8, or 12, 1-inch round square filters up to 5 mm thick. Second, third or higher order wavelengths of light diffracted from the grating can be eliminated by the use of a long pass filter.

Filter wheels can be manual or automatic. Automatic filter wheels are computer controlled, allowing the computer to switch filters during a wavelength scan to better isolate the desired wavelength.

8.5 Fiber-optic adapters

This adapter flange replaces any slit and can be mounted in the front or side exit ports. It can accommodate FC/PC connectors, NTT style.

8.6 Diode array mounting flange

This flange is mounted in the front of the exit slit. It can accommodate up to 2" diode array detectors. Its surface is tilted at an angle to optimize the flatness of the focal plane.

8.7 Light sources

Sciencetech offers arc and QTH lamps and power supplies for visible and near IR applications. The arc lamp housings for Xe, Hg-Xe or Hg lamps of up to 150W use a high efficiency ellipsoidal reflector with a N.A. matched to that of the spectrometer for maximum throughput and minimum stray light. For infrared applications, Sciencetech offers a chopper and infrared sources and detectors.

8.8 Detectors

Some of the possible accessories:

1. Single solid-state detector. Sciencetech provides flange mounted Si photodiode for the visible or UV enhanced Si for the UV/visible or Ge photodiode for near IR. The astigmatic segment length is approximately 5mm. This means that no light will be lost when using diode detectors of this diameter to measure.
2. PMT with housing. It bolts to the exit slit assembly.
3. A two-detector box is used with a manually operated folding mirror to switch detectors: a 5mm diameter Ge and a 6mm side Si photodiode, or for small signals, a PMT in place of the Si detector. There are internal optics to focus the input slit on each detector and a filter drawer in front of the Si detector to eliminate multiple diffraction orders from the grating. It accepts 1" round or square filters (Sciencetech offers square long pass filters from 320 to 1000nm), and it can also be used to hold interferometric 1" filters if desired. The detectors are mounted on a PC board with its own voltage regulators and transconductance preamplifier (10M Ω transconductance) for minimum noise in AC measurements. Ground loop pickup has been minimized. The signal cable from each detector is terminated with a BNC connector. The optics in the focus box has a magnification <1 in both the Ge and Si directions (our Ge detector has a diameter of 5mm). Light will start to be lost with a 5mm Ge detector when an input slit is used with a height larger than approximately 12mm, an extreme condition.
4. A 1" linear diode array mounts to side or front port

9 Installing A Folding Mirror (Model 9000-ST-X, X= In or Out)

The following procedure details how to install a Sciencetech model 9000-ST-IN folding mirror on the input of a 9055/9055/9057/9060 (and fast versions) monochromator. The procedure for a Model 9000-ST-OUT output folding mirror is the same, except the symmetry is opposite.

1. Remove the top lid of the monochromator. Remove the black baffle surrounding the grating turret via the three screws securing it.

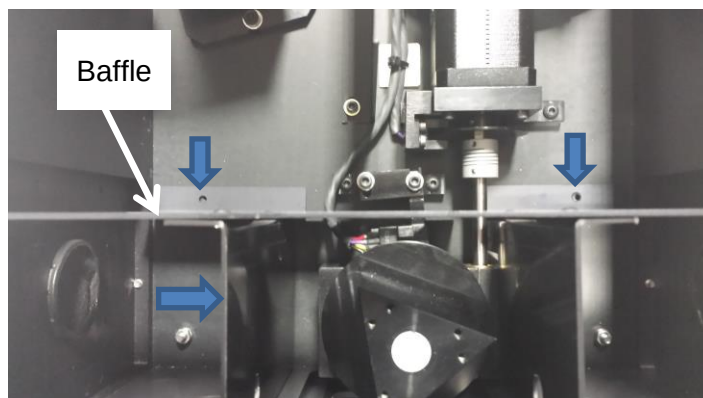


Figure 27. Removing black baffle around grating turret removing indicated screws.

2. This step is to setup the grating position that will allow final adjustment of the folding mirror after it is installed. Plug in the monochromator 24VDC supply and connect the USB to the computer. Start SciSpec (or SciUSBMono) and let the instrument initialize. After, unplug the 24VDC. Run a flat, straight HeNe or other laser directly into the front input port with a narrow-slit width. The beam should hit the centre of the first mirror at the proper optical height of 3". With a screen at the output port, and a slit width about the size of the laser beam, rotate by hand the motor shaft until the beam exits the exit slit and falls on the screen. Mark the position on the screen. This is the grating position for the last step. See Figure 28. Diagram of grating position below:

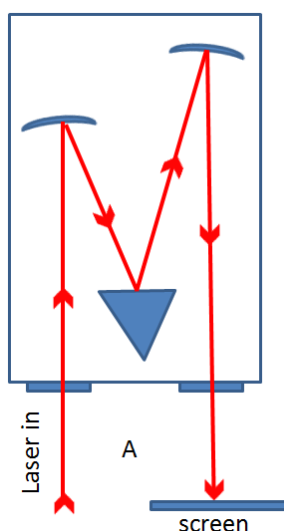


Figure 28. Diagram of grating position.

3. Remove the two M4 screws and their nuts and washers at the base of the input port. Be very careful of the gratings right in the vicinity of these bolts...wear powder free gloves. Poke out the black plastic cap to open the side input port.

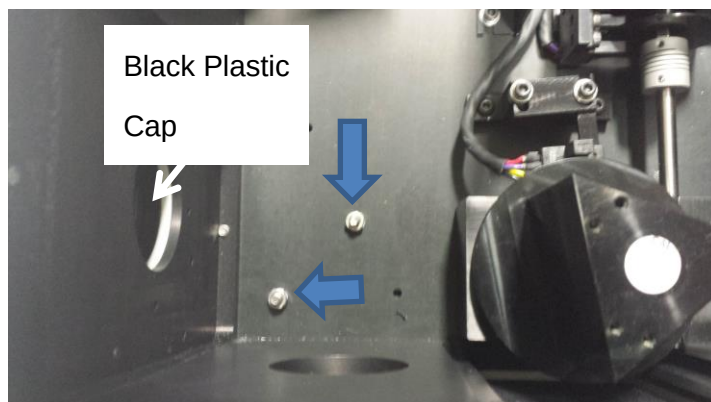


Figure 29. Blue arrows indicating M4 screws, nuts, and washers at input port base.

4. Insert the folding mirror into the monochromator and align the two mounting holes on the bottom with the two holes on the monochromator base plate. Secure it from the bottom with the two M4 screws.



Figure 30. Installed folding mirror into monochromator.

5. Take the slit off the front input port. Note the metal roll-pin on the backside of the slit that goes into a mating hole on the side black spacer plate. Be sure the O-Ring is mounted in its groove. Now mount the slit on the side input port spacer plate with the 4 screws provided. Take the black plastic plug removed from the side input port and install it in the now vacant front input port. Re-install the black baffle around the grating turret with the three screws.

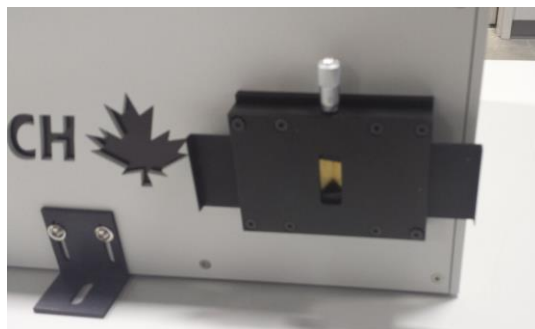


Figure 31. Mounted slit on the side input port.

6. To check the alignment of the folding mirror, set up the laser coming in flat and level normal to the new side entrance slit face. See Figure 32 below:

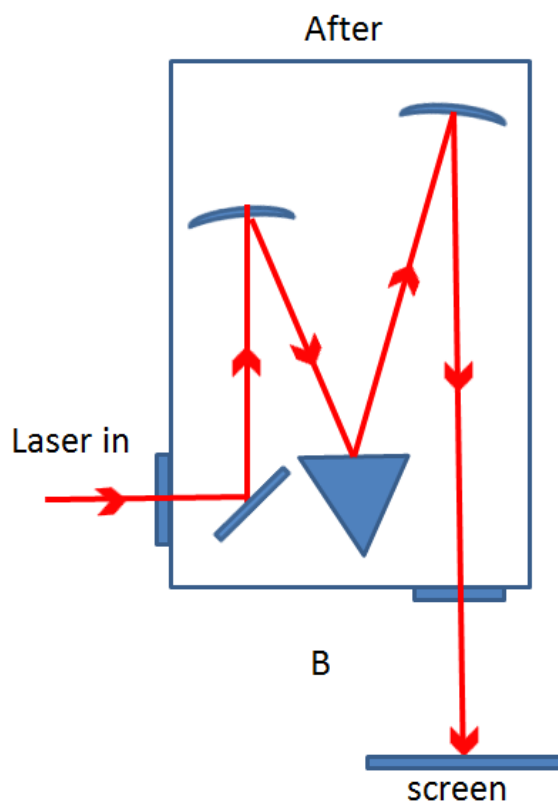


Figure 32. Diagram of path taken by light wave.

If the beam does not go through the exit slit to the same position on the screen as observed in Step 2, some adjustment to the folding mirror angle is required. There is an Allen head screw on the top of the folding mirror base, at the back of the mirror mount. Loosen this bolt and try to angle the beam so that it exits the output port to the screen. Re-tighten it if this works. Also, one can loosen the two M4 mounting bolts underneath the folding mirror, angle the mirror and then re-tighten them as well.

10 Available Gratings

10.1 Ruled Gratings

Grooves/mm	Blaze wavelength
50	12 or 18 μm
75	8, 10 or 12 μm
150	500 nm, 4, 6 or 8 μm
200	730 nm, 1.05, 1.7, 2.6 or 3 μm
300	300 or 500 nm, 2, 3, 3.5 or 4 μm
600	200, 300, 400, 500 or 750 nm, 1, 1.25, 1.6, 2.5 and 2.7 μm
1200	150, 200, 250, 300, 500, 600 or 750 nm, 1 or 1.2 μm
1800	500 nm
2400	300 nm

10.2 Holographic Gratings

Grooves/mm	Operation range
1200	190 - 800 nm
1800	200 - 800 nm
2400	200 - 600 nm
3600	200 - 400 nm

11 Dimensions

DISTANCE TO FOCAL PLANE	FROM SURFACE
0.125[3MM]	INSTRUMENT FACE (FRONT)
0.3125[8MM]	INSTRUMENT FACE (SIDE)
0.125[3MM]	SPACER (SIDE)
0.4375[11MM]	CAMERA CONFIGURATION (SIDE)

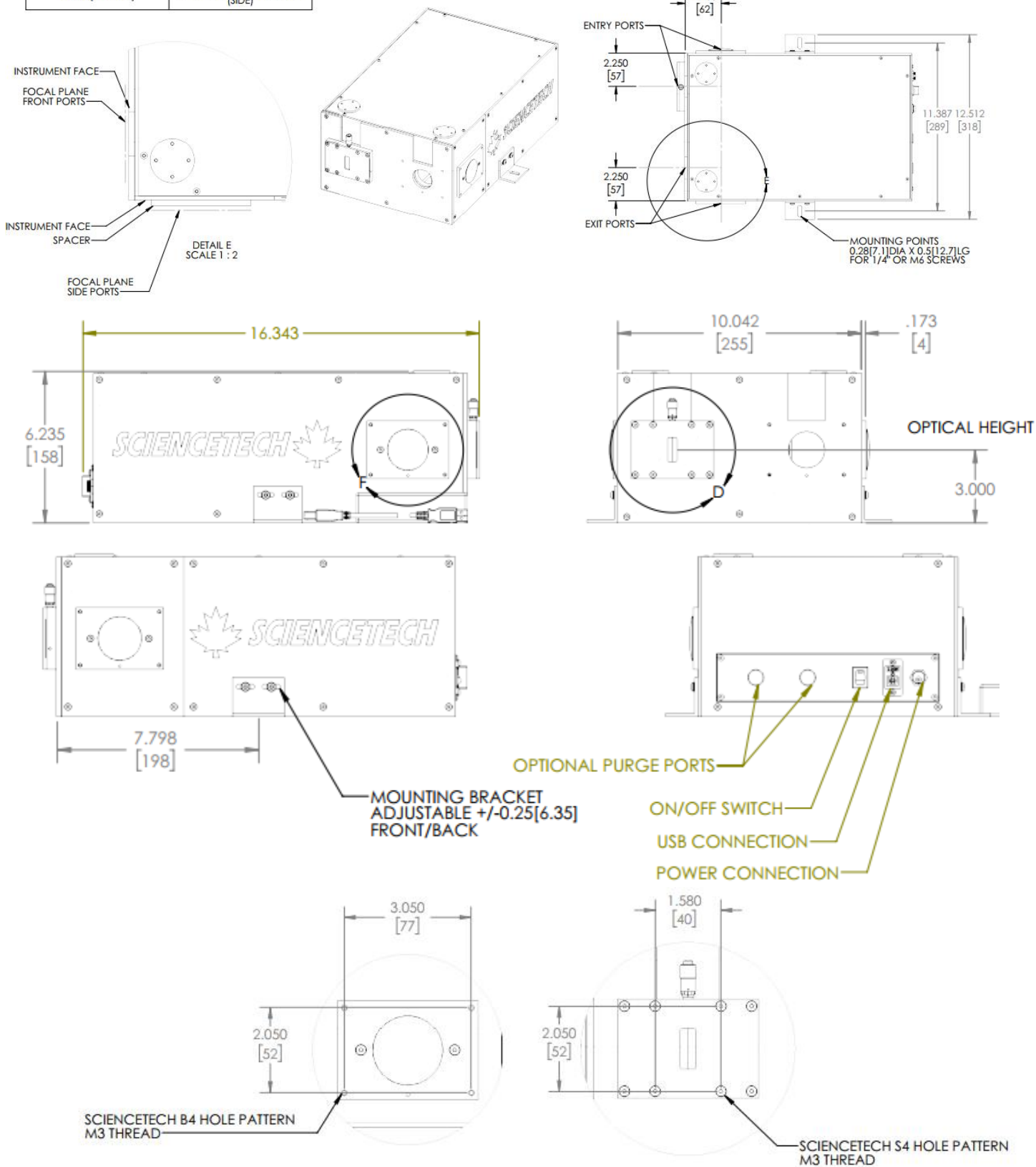


Figure 33: 9055 Dimensions.

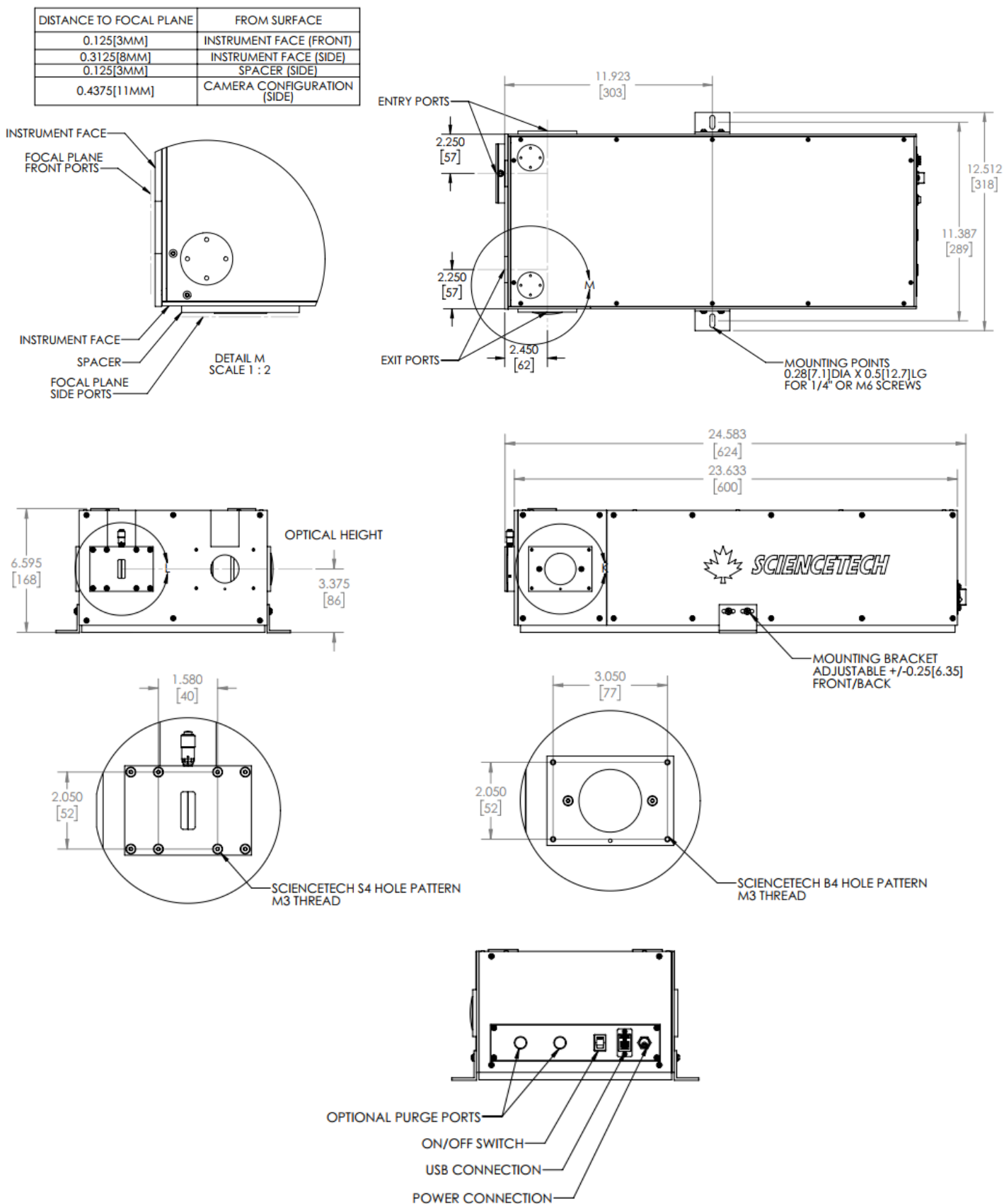


Figure 34: 9057 Dimensions.

12 Technical Specifications

Table 3. Specifications and Certificates of 9000 Series Direct Drive Monochromators

Physical Specifications				
	9055	9057	9055F	9057F
Drive	Computer controlled direct drive			
Number of Gratings (All with computer controlled grating selection)	Up to 3 in turret 50 mm square gratings		Up to 2 in turret 68 mm square gratings	
Input Mirror Focal Length	200 mm	457.2 mm	200 mm	457.2 mm
Output Mirror Focal Length	250 mm	457.2 mm	200 mm	457.2 mm
Aperture	f/3.5	f/7	f/2.5	f/6
Optical Resolution	0.2 nm	0.1 nm	0.4 nm	0.2 nm
Dispersion	4.0 nm/mm	1.8 nm/mm	3.3 nm/mm	1.8 nm/mm
Slit Height	Adjustable: 0-10 mm, 10-20 mm, or 20-30 mm			
Slit Width	Adjustable: 0.01-5.0 mm			
Height of Slit Center	76.2 mm (3.0 in)		86.4 mm (3.4 in)	
	11	22	11	22
Dimensions (mm)	289 x 158 x 396	264 x 162 x 611	289 x 158 x 396	264 x 162 x 611
Operating Environment	Indoor, well-ventilated environment. Not for household use.			
Operating Temperature Range	15 – 30 °C			
Operating Humidity Range	<80% RH, non-condensing			
Performance Specifications				
All specifications are for a 1200 l/mm grating except when stated otherwise.				
Wavelength Range	0 – 1000 nm (depending on grating)			
Wavelength Accuracy	±0.2 nm			
Wavelength Reproducibility	±0.1 nm			
Grating Change Reproducibility	±0.2 nm			
Electrical Specifications				
DC Input				
Voltage	24VDC			
Maximum Current	0.5 A			
Certificates				
CE Certification	9055 only. See Figure 35			



EU Declaration of Conformity

In accordance with European Parliament and Council Decision No 768/2008/EC

European Union Directives: ☒ 2014/30/EU Electromagnetic Compatibility Directive (EMCD);
☒ 2014/35/EU Low Voltage Directive (LVD);
☐ 2012/19/EU Waste Electrical and Electronic Equipment (WEEE); ☒ R&D Only;
☒ 2011/65/EU Restriction of Hazardous Substances Directive (RoHS); ☐ R&D Only;

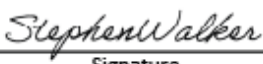
Harmonized Standards: ☒ EN 61236-1:2013 (EMC)(Note 1); ☐ Custom Permanent Installation (EMCD Clause 19.1);
☒ EN 61010-1:2010 (Safety);
☐ EN 62471:2008, IEC/TR 62471-2 (Photobiological Safety);

Manufacturer: Name: Sciencetech Inc.
Address: 1450 Global Drive, London, Ontario, Canada, N6N 1R3

Object of the declaration:
Equipment Type/Description: Computer Controlled Monochromator/Spectrograph with USB
Model Number(s): 9055 [120-9018]

Sciencetech Inc. declares its belief that the equipment listed above conforms to the selected European Union harmonized legislation. The listed equipment conforms to the selected directives and standards only if it is installed and operated in accordance with the manufacturer's specifications and instructions. This declaration of conformity is issued under the sole responsibility of the manufacturer.

Date: 20-Jan-2022



Signature
Stephen Walker

Full Name
V.P. Technology

Title

Note 1: Class A, Group 1

Figure 35. CE Declaration of Conformity

13 Important Notice

All electrical instruments may be dangerous if not handled in accordance with proper instructions and common precautions. Sciencetech Inc. will not be responsible for any damage caused by such units if instructions herein are not followed and repairs are not attended to or performed by company-trained or licensed personnel. All instruments should be operated with proper grounds on power line and should not be opened or handled as to electrical or electrically operated components without being switched off and disconnected from power receptacle.

Sciencetech Inc. reserves the right to make adjustments or improvements in its product without notice and without obligation to subsequent purchasers and without being required to make corresponding changes or improvements in products theretofore manufactured and sold.

We have done our very best in the manufacture and packing of this material. The transportation carrier is now responsible for delivering it to you in its original good condition, since all purchases are EXW London.

If the shipment is NOT delivered in good order and in accordance with quantity shown on Bill of Lading or Packing Slip, have the shortage or damage noted by the Carrier on both the delivery receipt and the freight bill, or by special form provided by United Parcel or the Post Office.

The Interstate Commerce Commission has ruled that Transportation Companies will not honor any losses or shortage claims unless exceptions are noted on the freight bill at the time of delivery. It is the buyer's responsibility to make a complete inspection immediately upon receipt of purchased goods.

If you accept shipment from the Transportation Carrier short of what is enumerated on the Bill of Lading – or in damaged condition – without having proper notation made by the Carrier, you do so at your own risk.

If bundles or crates are in apparent good order, but on opening contents are found to be damaged, call Carrier for adjuster to view same and have the Transportation Company/United Parcel/Post Office mark the freight bill or packing slip relative to such concealed damage. **Make your claim at once for the Transportation Company/United Parcel/Post Office has a limited time for presentation of claims.**

We are willing to assist you in every possible manner in collecting claims for loss or damage on this shipment, but this willingness on our part does not make us responsible for filing or collecting claims or replacing materials. Claims for Loss or Damage on shipment may not be deducted from our invoice, nor pay of the invoice withheld awaiting adjustment of such claims, as we cannot guarantee safe delivery.

Important: Do not return goods without written authority.

Contact factory for return material authorization.

Returned goods will not be accepted by us from the Transportation Company/United Parcel/Post Office unless written authorization has been issued by Sciencetech Inc.

Return of special or non-stock items cannot be authorized. Credit for goods returned - under authorization - will depend on the value to us based on our selling price, less a fair charge to cover the expense of shipping - re-handling - transportation - refinishing, etc., providing material is received in good condition - transportation charges prepaid - credit rendered to be used against future purchases.

All equipment manufactured by Sciencetech Inc. has been subjected to extensive performance and quality control testing. In order to constantly improve our product, we ask your assistance. Upon installation of our equipment, please fill out the attached card and return to us.

By completing the card and returning it to Sciencetech, you will register your instrument in warranty and enable us to provide you with the best possible service.

14 Warranty and Assistance

Sciencetech's warranty and return policy is subject to change.

If you have any questions or issues related to the operation of your 9000 series monochromator and its accessories, please contact:

Sciencetech Inc.
1450 Global Dr.
London, ON N6N 1R3, Canada
Telephone: 519-644-0135
Fax: 519-644-0136
Email: sales@sciencetech-inc.com

For our up-to-date contact and warranty information please see:

<http://www.sciencetech-inc.com/warranty-returns>

If your shipment is received in damaged or poor condition, please see Sciencetech's detailed policies on how to proceed on the Sciencetech website